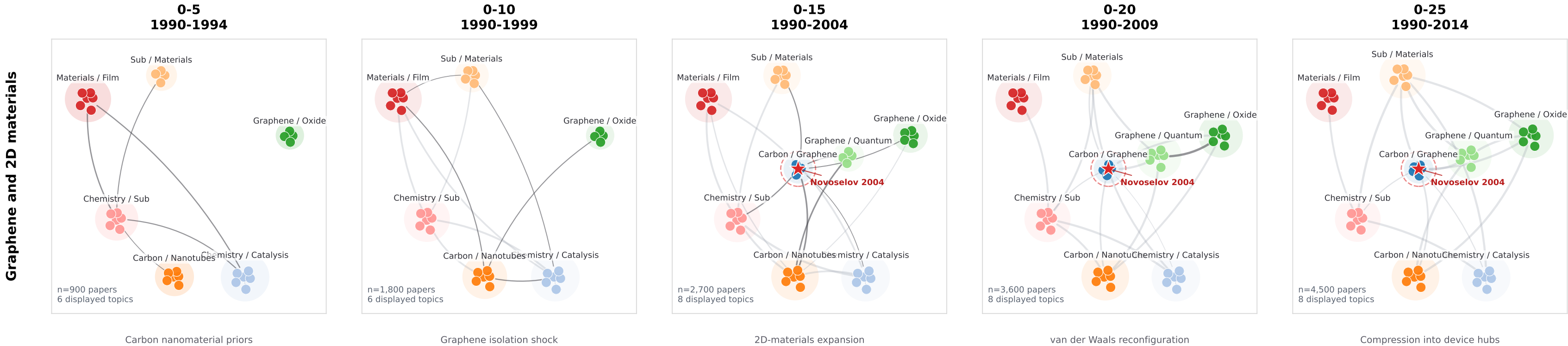


Fig. 1 | Citation knowledge-graph perturbation after a landmark innovation

Example domain: Graphene and 2D materials. Real OpenAlex data; publication-style display uses data-derived topic snapshots.



a Cumulative citation knowledge graph snapshots



b Perturbation fingerprint across rolling 5-year intervals

